

Introduction

Table: Comparing how process development/biomanufacturing considerations are different for in-space biopharmaceutical manufacturing vs. current considerations on Earth. For example:

- Scale: Large scale on Earth; in space, need small batches and use the same equipment to produce different pharmaceuticals
- Process optimization: on Earth, optimize to maximize yield and reduce costs; in space, the “costs” are related to upmass, so optimize to minimize mass, power, volume, crew time
- Reagent storage: cold storage widely available on Earth; in space, cold storage very limited
- Regulatory: established requirements by regulatory agencies on Earth; in space, currently undefined
 - Manufacturing environment: cGMP production on Earth; in space, no considerations yet for how to translate GMP principles (if at all)
 - Quality control
 - In-human trials of produced biopharmaceuticals

Methods

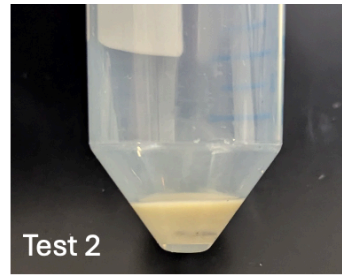
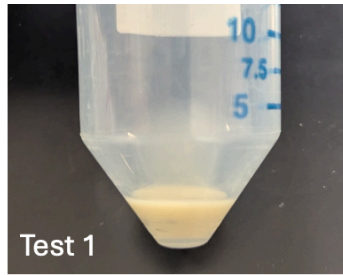
Figure: Block diagram outlining required process steps, inputs/outputs for each step, flow between steps

Figure: pictures of hardware used for prototype testing experiment

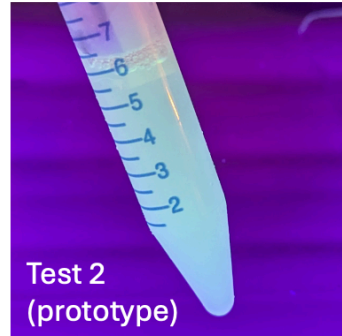
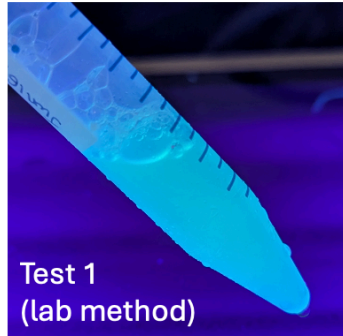
Results

Figure: showing sensor data over time

Cell Pellets



**Soluble
Lysate
Fraction**



**Insoluble
Lysate
Fraction**



Figure: Showing how much culture we got in both test, presence of GFP-tagged peptide in soluble lysate fraction; still some GFP may be left behind in insoluble lysate fraction (either due to formation of inclusion bodies or insufficient lysis)

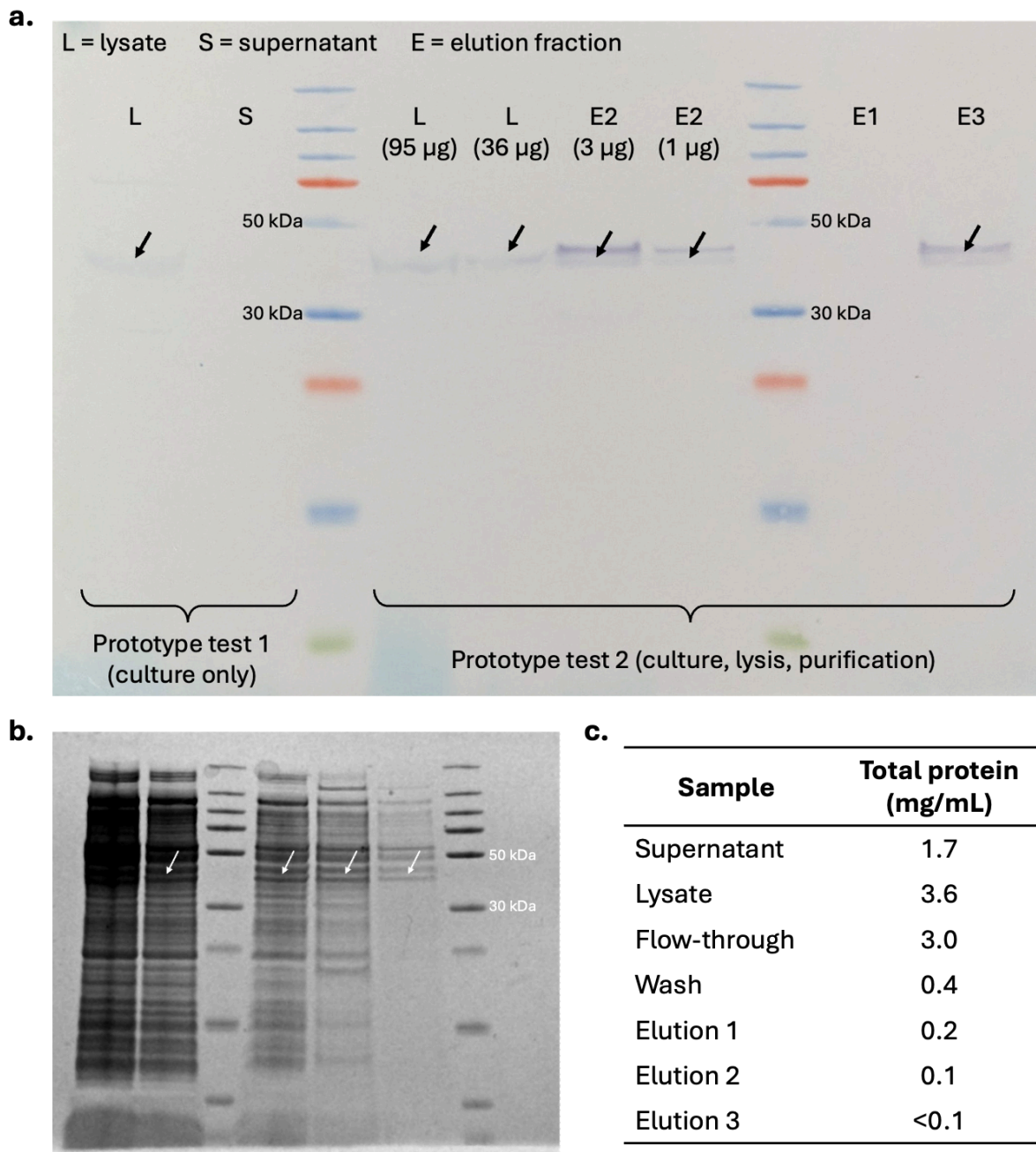


Figure: Western Blot, protein gel with total protein staining, BCA assay results

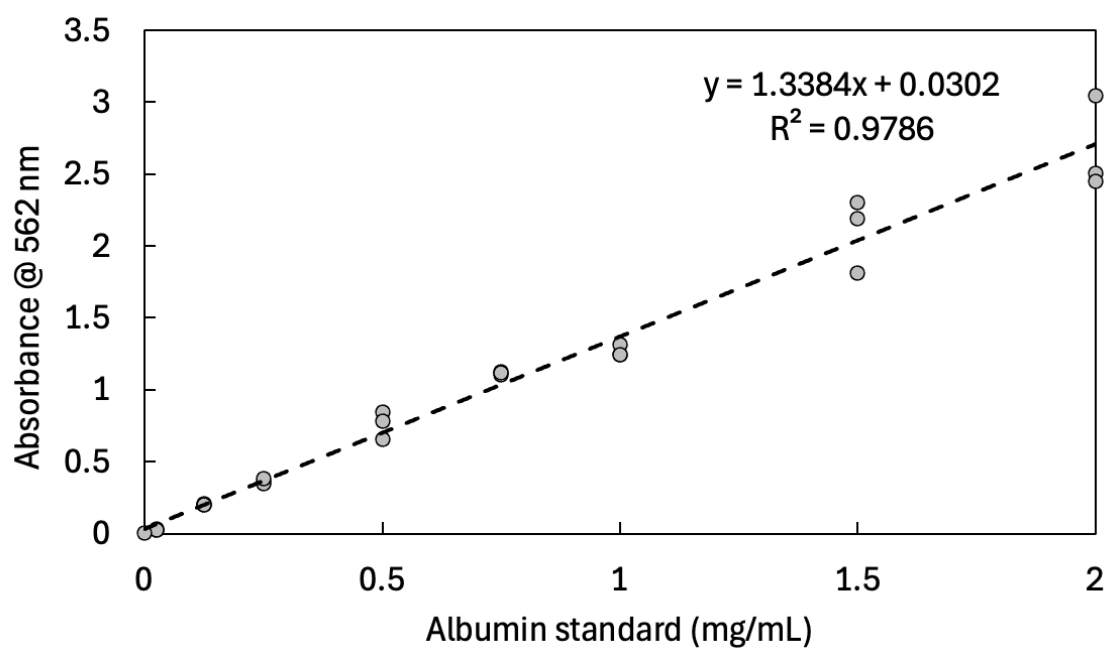
Table: mass, power, volume for prototypes used in this study

Discussion

Figure: maybe roadmap to further prototype development & implementation of Astropharmacy? E.g., next steps in prototype development and testing of the entire system, identifying process control/sensor needs, which reagents need to be tested for stability at RT, thoughts about quality control, etc.

Could also be the roadmap that we sent to MCO...

Supplementary Figures



BCA assay standard curve

Western blot standard curve & calculations